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Honors Thesis:
Digital Pollution: Understanding the Energy Costs of Large Language Models

Introduction

In an age when our lives are increasingly digitized, the environmental costs of our digital habits often go unseen. Every email, photo, cloud file, and now interaction with large language models (LLMs) relies on physical infrastructure. Behind every query are data centers filled with servers that consume enormous amounts of electricity and water to train, store, and generate information. Scholars refer to this invisible demand for power as digital pollution [1]. But the term itself requires careful unpacking.

Digital pollution is best understood as an umbrella concept covering the full lifecycle impact of digital technologies, including e-waste, hardware manufacturing, and the long-term energy demands of data infrastructure [18]. Within this broader category, a more specific and frequently underappreciated phenomenon is Internet Pollution: the carbon and energy footprint of active data transmission. This includes streaming video, cloud uploads, and, increasingly, queries to large language models. The distinction matters for this research because they clarify which costs are fixed (e.g., the carbon embedded in manufacturing a GPU) and which scale with user behavior (e.g., the energy cost of each AI response). Both layers of digital pollution deserve attention, but user behavior has the most direct potential for mitigation through awareness and education.

At the same time, technology is accelerating discovery across science, education, and industry. As computational tools become more powerful, they also introduce a growing responsibility to use them effectively and thoughtfully. This research approaches AI not as a technology to reject, but as one to understand better and use responsibly.

The goal of this project is to measurably improve public understanding of sustainable digital usage and how large language models consume energy and resources, and to evaluate through surveys whether targeted education can influence users' attitudes and intended behaviors. To achieve this, the research adopts a creative approach that combines empirical analysis with an interactive educational website designed to visualize and explain ways to prevent e-waste and the energy costs associated with LLM training and inference.

The central research question guiding this study is: Do people understand the energy and emissions costs associated with large language models, and can a brief educational intervention improve that understanding and shift behavioral intentions?

To address this question, the study focuses on the following variables:

- Objective knowledge: what users actually know
- Perceived knowledge: what users believe they understand
- Attitudes: how important sustainability and transparency are to users
- Behavioral intentions: how users plan to act after learning
- Exposure: whether users interacted with the educational website (pre vs. post)

Through surveys, data analysis, and educational design, this study evaluates how awareness of LLM energy use shapes user understanding and behavior.

Background

Sustainability has been a personal value of mine for as long as I can remember. I grew up in Arizona and California during the 2010s and experienced firsthand the impact of drought and wildfire. Daily conservation practices, including reusing water by washing clothes while showering and minimizing waste by saving rinsing water from produce for reuse in gardening, were part of my upbringing. These early experiences and being raised by a Buddhist family taught me the importance of conservation, respect for wildlife, and interconnectedness, in which all humans, animals, and nature are profoundly linked, and damaging one part harms the whole. These childhood teachings are my primary motivation for understanding how technological progress can align with ethical and environmental responsibility.

I approach this research from a position that values both technological acceleration and environmental responsibility. As a Computer Science major with a background in data science and marketing, I have seen how computation can dramatically accelerate discovery across science and engineering. At the same time, through conversations with scientists and engineers, I became aware of the vast physical infrastructure required to support digital systems, including energy-intensive data centers and large-scale storage facilities. In many contexts, these material costs remain secondary to performance and scale.

Bridging Research and Usability in Renewable Energy Systems

In 2023, I was first introduced to a renewable energy initiative within the Grid Integration Systems and Mobility (GISMo) group at SLAC National Accelerator Laboratory. I received guidance on translating complex technical systems into accessible resources for researchers and utility developers through website development and Git-based code management. Much of my early work focused on bridging gaps between research and usability. I led publication efforts for multiple energy and sustainability initiatives, creating platforms that shared training materials, simulation tools, and strategic research outputs. [12]

Through each project, I became increasingly exposed to the hidden complexity behind the scenes—version control, Git, and so on—but that was just the beginning. Even as these platforms expanded access to cutting-edge research, the physical systems supporting them, servers, computation, energy use, and cooling, were rarely part of the conversation.

From Visualization to Grid Resilience under Extreme Weather

The importance of energy efficiency became more concrete during my work on the Renewable Energy Generation Risk from Outlier Weather (REGROW) project at SLAC National Accelerator Laboratory [10]. REGROW focuses on developing reproducible methodologies and data-driven tools to assess vulnerabilities and improve the performance of photovoltaic systems during extreme weather events, to enhance electrical grid resilience.

Using a reactive Python notebook environment, *marimo* [15], I analyzed meteorological and energy data from the August 2020 California heat wave. I integrated high-resolution data on temperature, solar irradiance, and wind speed to examine anomalous weather patterns and their effects on renewable energy generation. I built interactive visualizations that served as ground truth for comparing outputs from risk simulation models and optimization strategies. These analyses highlighted the importance of careful system-level optimization, including battery charging and discharging strategies, for maintaining grid stability during periods of high demand.

Computational Efficiency and Responsibility in Scientific Discovery

In 2025, my Science Undergraduate Laboratory Internship (SULI) research at the Stanford Synchrotron Radiation Lightsource (SSRL) pushed this perspective by putting computational efficiency at the center of scientific discovery. I explored methods to automate beamline experiments by integrating real-time analysis with beamline control systems. Using Python-based automation and a two-detector configuration, the system analyzed fast, low-resolution diffraction data and triggered high-resolution scans only in information-rich regions near diffraction peaks.

Applied to the anatase-to-rutile phase transformation in titanium dioxide (TiO_2), this approach reduced unnecessary background scans and improved time efficiency while preserving data quality for quantitative analysis. This experience demonstrated that thoughtful system design can improve scientific throughput while conserving limited resources, such as beam time, computational resources, and energy.

Automation and machine learning are often discussed only in terms of speed and scale, yet this project demonstrated that thoughtful system design can also promote efficiency and sustainability. Rather than indiscriminately maximizing data collection, the system prioritized informative measurements, illustrating how intelligent automation can improve scientific throughput while conserving limited resources, such as beam time, computational resources, and energy. This experience marked a turning point in how I think about responsibility in computing, as decisions about when to collect data, how much to compute, and which processes to automate carry real environmental implications.

Together, these experiences shape the motivation for this honors thesis. I approach large language models not as technologies to oppose, nor as tools to adopt uncritically, but as systems whose benefits must balance against their growing energy and infrastructure demands. This thesis reflects a search for a middle ground, one that preserves AI's capacity to accelerate discovery while emphasizing transparent, efficient, and sustainable use.

Literature Review

Digital pollution remains underrepresented in broader environmental discussions, despite the growing material footprint of digital technologies. Every interaction with a large language model depends on physical infrastructure, including data centers, servers, electricity grids, and cooling systems [1]. As AI systems become deeply embedded in scientific research, business workflows, and everyday productivity tools, their environmental and economic impacts warrant closer examination.

Digital pollution remains underrepresented in broader environmental discussions, despite the growing material footprint of digital technologies. This sustainability topic serves as an umbrella concept encompassing both the static costs embedded in hardware manufacturing and e-waste, and the dynamic costs of active Internet use, which is increasingly referred to as Internet Pollution [18]. Every interaction with a large language model belongs to this second category, drawing electricity from grids and water from cooling systems with each response [1]. As AI systems become deeply embedded in scientific research, business workflows, and everyday productivity tools, their environmental and economic impacts warrant closer examination.

Digital Pollution and Consumer Mitigation Strategies

To understand where digital pollution originates and how individuals can reduce their contribution, it is useful to examine specific everyday behaviors and their associated costs. According to Greenly, common sources of digital pollution include storing undeleted emails, streaming video through cellular networks rather than Wi-Fi, enabling automatic video play on platforms like YouTube, subscribing to unnecessary cloud storage plans, and refreshing social media feeds repeatedly throughout the day [18]. Each of these activities places additional load on server infrastructure, triggering energy consumption that is invisible to the end user but physically real at the data center.

Internet pollution, a subset of digital pollution, compounds these individual behaviors at scale. A single unread email stored in a server contributes a small but non-zero carbon cost; multiplied across billions of accounts worldwide, persistent digital clutter becomes a meaningful source of emissions. With approximately 2.3 billion people currently using cloud storage services, the strain on data centers continues to grow, making local storage alternatives—such as hard drives or solid state drives—a meaningful way to reduce server energy demand [18]. Choosing Wi-Fi over mobile data (4G/5G) for internet-intensive tasks can meaningfully reduce the energy cost of data transmission, since mobile networks are generally less energy-efficient per bit transmitted than fixed broadband infrastructure [18].

For consumers seeking to reduce their digital footprint, several practical behavioral changes are supported by the literature. Deleting old emails and attachments, unsubscribing from unused newsletters, disabling autoplay features on streaming platforms, favoring Wi-Fi over cellular data, and keeping devices for longer periods to reduce manufacturing-related e-waste are all recommended mitigation strategies [18]. In the context of AI specifically, analogous behaviors include writing concise, targeted prompts rather than open-ended ones; batching related questions into a single query; and selecting smaller, task-appropriate

models rather than defaulting to the largest available option [4], [6]. These consumer-level actions form the behavioral foundation that the educational website component of this thesis aims to promote.

Training and Inference in Large Language Models

To ground this discussion, it is important to define the term "large language model" (LLM). Large language models are machine learning systems trained on massive text datasets to generate human-like language by predicting word sequences. Two computational processes dominate their energy use: training and inference. Training involves adjusting billions of parameters through repeated computation on large datasets, often requiring hundreds or thousands of GPUs or TPUs to operate continuously for weeks or months. Researchers estimate that training a single state-of-the-art LLM can consume megawatt-hours to gigawatt-hours of electricity, comparable to the annual energy use of hundreds of U.S. households [3], [5].

Inference, by contrast, refers to generating outputs after deployment. While a single inference request requires far less energy than training, inference becomes a major contributor to long-term energy demand when scaled to millions of daily interactions worldwide [6]. Because measurement practices vary and reporting remains inconsistent, non-technical users often misunderstand the distinction between these processes [4], [7].

Response Verbosity and the Energy Cost of Every Token

A particularly underappreciated driver of inference energy use is response verbosity, the length and detail of AI-generated outputs. Research in human-computer interaction consistently finds that users prefer longer, more elaborated AI responses, perceiving them as more helpful and authoritative. However, from an energy perspective, every additional word in a model's reply represents one or more additional "forward passes" through the GPU: each output token requires the model to execute a full computation through its neural network layers. Longer responses, therefore, scale energy consumption nearly linearly with output length.

This dynamic is further amplified by a new class of models known as "Thinking" or "Reasoning" models, such as Qwen 3 Thinking and similar architectures, that generate extended internal chains of thought before producing a final answer. These internal reasoning steps are themselves token-generating processes, meaning that a single user query may trigger tens of thousands of output tokens before the visible response is ever produced. Compared to a standard chat model, reasoning models can consume energy at an order of magnitude greater per request, making model selection a consequential energy decision for users and developers alike.

This dynamic creates a structural tension in AI product design. Optimizing for user satisfaction by generating verbose responses directly increases computational cost and energy demand at scale. When millions of users interact with a model daily, the aggregate effect of preferring longer answers over concise ones represents a meaningful and largely invisible driver of data center energy consumption. Raising awareness of this relationship, that a shorter, well-scoped prompt can reduce the energy footprint of a response, is one practical behavior change this study aims to promote.

The GPU Bottleneck: Hardware Efficiency and the Efficiency Paradox

The energy cost of a single inference token is not fixed; it depends substantially on hardware generation and model architecture. Newer GPU architectures, such as the NVIDIA H100, are considerably more energy-efficient per floating-point operation than older accelerators like the A100, offering improved performance-per-watt ratios. However, this hardware efficiency gain does not automatically translate into lower total energy consumption. Because more capable hardware enables the deployment of more complex and parameter-rich models, including the "reasoning" architectures described above, the total energy cost per user request frequently increases even as the underlying silicon becomes more efficient.

This phenomenon, sometimes called the Jevons paradox in energy economics, means that measuring AI sustainability requires examining full system-level costs rather than hardware efficiency alone. The ML.ENERGY Leaderboard's direct measurement approach captures this distinction by recording actual power draw during real inference workloads rather than relying on theoretical hardware specifications.

Comparing AI to Conventional Search: A Meaningful Benchmark

To contextualize the energy demands of generative AI, it is useful to compare them with conventional web search, the technology that AI tools are increasingly being asked to replace or supplement. A single Google search query consumes roughly the energy equivalent of running a 60-watt lightbulb for 17 seconds. Generative AI responses, which require significantly more computation, can consume substantially more energy depending on the size of the model and the length of the response. Importantly, this comparison is not fixed: smaller, more efficient models can narrow this gap considerably, while large frontier models widen it.

This comparison matters for public understanding because it offers a relatable reference point. Most users have some intuition that using the internet consumes energy; far fewer understand that replacing a search with an AI-generated response can represent a meaningfully larger energy cost per interaction. Educational interventions that anchor AI energy use to familiar comparisons like lightbulb-seconds or household equivalents have the potential to make abstract statistics personally meaningful.

Ethical Foundations and Transparency

As large language models (LLMs) have grown in size and capability, so too has their energy consumption. To address the knowledge gap, the educational component of this thesis focuses on providing universally understandable explanations of how large language models process prompts, where energy costs arise, and how users can adopt lower-energy alternatives.

Training a single LLM can require extensive computational resources, often involving thousands of high-performance accelerators operating simultaneously and consuming electricity frequently sourced from non-renewable grids [18]. This high energy demand has several implications. From an

environmental perspective, data centers that support AI workloads contribute directly to greenhouse gas emissions, particularly in regions where electricity generation remains dependent on fossil fuels [4]. From a water-use perspective, cooling systems for AI clusters can consume millions to tens of millions of liters of water per year, an impact that is often overlooked in popular discussions of AI sustainability [4]. In water-stressed regions such as California, this level of consumption is comparable to the annual water required to produce thousands of pounds of almonds, a crop frequently cited for its high irrigation demands. Economically, the cost of powering and maintaining these systems can reach millions of dollars per year, influencing which organizations can afford to develop advanced models and raising ethical questions about energy allocation and access [8].

These concerns align with early ethical critiques of large-scale language models. In *On the Dangers of Stochastic Parrots*, Gebru et al. argue that the dominant research culture rewards scale and performance while sidelining environmental and social costs [2]. Although the paper does not directly measure energy consumption, it emphasizes that limited transparency around training data, compute resources, and system design prevents meaningful accountability. This critique provides a foundational rationale for examining not only how much energy AI systems consume, but also how that information is communicated to users, policymakers, and decision-makers.

Measuring Energy Consumption and Emissions in LLMs

Measuring the energy and emissions burden of LLMs remains a significant methodological challenge, and one that is frequently distorted by a common misconception: the TDP Myth. Thermal Design Power (TDP) is the maximum heat a GPU is designed to dissipate, often conflated with maximum power consumption. Many informal energy estimates for AI systems calculate cost by multiplying a GPU's rated TDP by runtime, assuming the hardware is always operating at its thermal ceiling.

However, real-world benchmarking data from the ML.ENERGY Leaderboard project, which directly measures GPU power draw during LLM inference using hardware power sensors, shows that actual power consumption typically ranges from 25 to 75 percent of rated TDP, depending on model architecture, batch size, and workload type [17]. In practical terms, TDP-based estimates can overstate actual energy use by up to a factor of four. This systematic overestimation has consequences in both directions: it can alarm the public with inflated figures while simultaneously making efficiency gains look smaller than they are.

The ML.ENERGY Leaderboard addresses this gap by providing direct, hardware-level power measurements for inference tasks across a range of open-source models, reported in joules per output token [17]. For reference, the benchmark shows that a large model such as DeepSeek R1 consumes approximately 2.37 joules per token, whereas a smaller model such as GPT OSS 20B uses only around 0.028 joules per token. This is a difference of nearly two orders of magnitude that illustrates how dramatically model selection affects energy footprint. These measurements offer a reproducible, empirically grounded basis for comparison, one that this thesis adopts for its own benchmarking experiments.

Analyses summarized by MIT Technology Review further explain that credible estimates must specify their system boundaries, such as whether calculations include only accelerator power or encompass full data-center operations, including cooling, redundancy, and power distribution [4]. Because many corporate disclosures remain opaque, ranges and uncertainty intervals are more honest than single-point estimates.

Despite measurement-level uncertainty, large-scale trends consistently indicate that AI workloads are becoming a dominant driver of data center growth. Reporting by WIRED suggests that AI may account for up to 20 percent of global data-center electricity demand, with projections indicating substantial increases by the end of 2025 [8]. Although experts caution that precise attribution remains difficult due to limited disclosure, the direction of growth is clear.

Drivers of LLM Energy Demand

Several factors influence the energy consumption of LLMs. Model size plays a central role, as larger models require more computational operations and memory movement during both training and inference, increasing overall power demand [6]. Computational resources, including the number and type of GPUs or TPUs, directly affect power draw, with modern accelerators consuming hundreds of watts per device under load [5]. Training duration, infrastructure efficiency, algorithmic design, hyperparameter tuning, and data preprocessing also contribute to total energy use.

The IEEE literature increasingly emphasizes that algorithmic efficiency and system-level optimization can reduce unnecessary computation, suggesting that sustainability is not solely a function of scale but also of design choices [6]. Benchmarking studies clarify what "energy per use" entails at the level of individual interactions. From Words to Watts reports inference costs on the order of a few joules per output token, with total power draw varying by model size, batch configuration, and hardware type [6]. These results further demonstrate that meaningful comparisons require standardized experimental settings and clearly reported assumptions.

Model size remains one of the most direct drivers of energy demand. Smaller open-source models with single-digit billions of parameters can often run on a single GPU. In comparison, larger models with tens of billions of parameters require multiple GPUs or specialized serving infrastructure. This scaling multiplies both electricity consumption and cooling requirements [5], [6].

Public Understanding and Misconceptions

While technical discussions often focus on infrastructure and algorithms, public understanding of AI’s environmental impact remains limited. Over the past decade, information on AI applications has expanded rapidly. As documented in *Artificial Intelligence: A Modern Approach* and the AI Index Report, publications in machine learning increased by more than twentyfold, student enrollment in AI-related courses rose sixteenfold internationally, and conference attendance at venues such as NeurIPS increased by approximately eight hundred percent [9], [14]. AI has surpassed early expectations, enabling advances

in autonomous vehicles, machine translation, speech recognition, medical diagnostics, and climate science.

Alongside these benefits, however, significant environmental risks persist. While AI has the potential to reduce repetitive labor and accelerate scientific discovery, limited disclosure around training resources and infrastructure continues to prevent meaningful accountability [2]. Recent analyses indicate that approximately seventy percent of new AI-related articles present neutral or positive narratives, with ethical discussions often focusing on data privacy and algorithmic bias rather than environmental impact [14]. At the same time, diversity within AI research remains skewed, with men comprising roughly eighty percent of faculty and industry leadership positions worldwide, a pattern mirrored among doctoral students and technical hires [14]. These trends underscore the importance of examining not only how much energy AI systems consume, but also how that information is communicated beyond technical communities.

Survey data from the Searchlight Institute further illustrate this gap. Polling reveals that many respondents underestimate the energy required to operate AI systems and frequently conflate training with inference [7]. In some surveys, fewer than half of respondents correctly identify data centers as major contributors to AI's environmental footprint, with individuals from non-technical backgrounds particularly likely to misattribute impacts. At the same time, the survey finds that support for transparency and regulation increases markedly when respondents are informed about AI's energy and water use, suggesting that misinformation stems less from apathy than from limited exposure to clear explanations [7]. Recommended practices include reducing unnecessary prompts, batching tasks, and selecting smaller or more efficient models when possible [4], [6].

These infrastructure-driven impacts align with economic theories of externalities, in which private actors benefit from computation while distributing environmental costs across society. As Pigou's welfare economics framework explains, such unpriced external costs often lead to overconsumption in the absence of regulation or transparency mechanisms [16].

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Media narratives further reinforce widespread misconceptions by framing the cloud as intangible. As *MIT Technology Review* observes, "people think that the cloud is just an invisible thing, when in reality it's very physical with cables and energy centers to store everything" [4]. This disconnect between user experience and physical infrastructure is especially relevant for everyday users and business owners who

rely on AI tools to increase productivity without seeing how usage patterns contribute to cumulative energy demand.

Implications for Research Design and Hypotheses

Together, this literature helps pull back the curtain on how AI systems actually work and why their environmental impact deserves attention. Each prompt and response does not emerge from thin air; it travels through warehouses of machines, humming servers, power lines, and cooling systems that quietly draw from finite resources. The gap between what users *think* they understand about AI's energy use and what they actually know motivates this study's focus on both objective and perceived knowledge. Research demonstrating that awareness can increase support for transparency and regulation informs hypotheses regarding attitudes toward sustainable AI practices. Similarly, evidence that everyday usage patterns drive cumulative energy demand motivates a focus on behavioral intentions, such as reducing unnecessary queries or repeated prompts.

In summary, existing research shows that the environmental impact of LLMs is real, multifaceted, and often hidden in plain sight. While precise measurement remains contested, there is broad agreement that AI systems impose meaningful energy, water, and economic costs. Notably, the literature suggests that sustainability is shaped not only by technological design but also by how it is understood and used. This work frames each interaction with AI as a small but consequential choice, inviting users to engage with technology more mindfully and with care for the natural world that ultimately powers every response.

Methodology

This project integrates three methodological components:

1. Data Analysis: Compile and compare published energy benchmarks from open-source models such as BERT, GPT-NeoX, and LLaMA to estimate carbon footprints across architectures, using direct hardware measurement data from the ML.ENERGY Leaderboard rather than TDP-based estimation.
2. ~~Survey Component: Conduct pre- and post-education (1–2 weeks later) surveys among students and working professionals to measure awareness of digital pollution and AI sustainability.~~
3. Prototype Development: Build an interactive and educational website using HTML, CSS, and JavaScript that visually simulates how a model processes a prompt, tracks its energy use in real time, and provides accessible comparisons to help users internalize the cost of their AI interactions.

Survey and Interview Design

This study will collect data from two primary populations: (1) everyday AI users, including teenagers, college students, and working adults, and (2) data scientists and AI engineers at large technology

companies. These groups represent the consumer-facing and technical sides of AI energy consumption, allowing comparison between public understanding and expert perspectives.

Population of Interest

General AI Users include college students and working adults who use AI tools such as ChatGPT, Google Gemini, Copilot, or any LLM-powered service. This group comprises everyday users who contribute to the growth of digital energy demand.

Technical Experts include data scientists, machine learning engineers, cloud infrastructure specialists, and AI researchers at large technology companies (Google, Microsoft, Amazon, Meta). These participants provide insight into how models are trained, deployed, optimized, and maintained.

Recruitment Process

Participants will be recruited after IRB approval through Pace University student groups and the researcher's professional network. Student participants (general AI users) will be recruited via brief announcements or digital flyers shared in Seidenberg clubs and communities, including the AI Club, Women in Tech, Google Developer Student Club, Software Engineering groups, Algorithms classmates, Business Analytics peers, and at Seidenberg events with permission from organizers. The study link may also be shared on the researcher's LinkedIn to reach Pace students and alums.

Technical experts will be recruited through existing professional connections, including industry contacts, guest speakers, and colleagues from prior events or internships. Recruitment will occur through email or LinkedIn messages.

All recruitment materials will include the study purpose, voluntary nature of participation, eligibility criteria, and a link to the online consent form. No private email lists will be used.

Constructs Measured

The survey will measure the following constructs using short Likert-type scales (5- or 7-point scales):

Knowledge: awareness of how LLMs consume electricity and water; understanding of the difference between training and inference; understanding of how response length and model selection scale energy use

Attitudes: beliefs about the environmental impact of digital technologies; perceptions of responsibility (individual vs. corporate)

Behaviors: frequency of AI use; efforts to limit unnecessary computational demand

Intentions: willingness to adopt more sustainable digital habits; support for transparency in AI energy reporting

Pretest–Posttest Design (Quasi-Experimental)

Participants will complete a pretest measuring baseline knowledge and misconceptions. They will then complete the educational intervention by exploring the interactive website that explains LLM energy use. A posttest will then measure changes in understanding, awareness, and attitudes. A statistically significant difference between the pretest and posttest indicates improvement and supports rejection of the null hypothesis.

Expert Interviews

To complement the survey, the study will include in-depth interviews with data scientists, AI engineers, and cloud specialists. These semi-structured interviews will explore: the true energy costs of training and deploying LLMs; optimization strategies used by large tech companies; barriers to implementing energy-efficient AI; public misconceptions about AI energy use; and how companies measure and report environmental impact.

Prototype Development and Experimental Methods

The experimental component of this thesis combines empirical measurement with interactive design to make the energy costs of large language models visible and understandable. I will benchmark the energy use of representative open-source models by conducting controlled inference experiments on cloud-based GPU infrastructure, enabling standardized, reproducible measurement of computational energy consumption across model sizes and hardware configurations. Critically, measurements will be recorded using direct hardware power sensors rather than TDP-based estimates, in line with the ML methodology.ENERGY Leaderboard [17], to ensure accuracy and avoid the systematic overestimation that characterizes many informal analyses.

These experiments are supported by awarded research funding, which enables access to managed cloud resources and built-in energy and carbon monitoring tools. The Zeus library is the open-source measurement framework that powers the ML.ENERGY benchmark will be used to instrument GPU power draw during inference, providing a reproducible and community-validated measurement methodology.

The proposed website is designed as an interactive platform that helps users understand the infrastructure, energy requirements, and financial costs behind AI systems. Rather than only explaining digital pollution, the site will allow companies, developers, and researchers to simulate how different AI workloads translate into real-world computing resources.

Users will be able to input parameters such as the type of AI model, the number of users interacting with the system, the number of prompts processed, and the type of hardware used (for example, CPU or GPU configurations). The platform will then estimate the energy required to generate those outputs, how long the system would take to process them, the number of machines needed to support the workload, and the approximate operational costs.

For example, an AI company could explore questions such as:

- How much energy is required to generate a certain number of responses?
- How many users can be supported on a given processor or server cluster?
- How many prompts can be processed before additional infrastructure is needed?
- How long does it take to power up and run the system continuously?
- What are the financial costs of running this system at scale?

The platform will also connect these infrastructure needs with potential solutions. Based on the simulated workload, the site can suggest more efficient hardware options, such as lower-energy CPUs or AI accelerators, or provide information about cloud providers and data centers that rely on renewable energy sources. In this way, the website acts as a bridge between companies building AI products and the developers, hardware providers, and infrastructure partners who design more energy-efficient systems.

Beyond simulation, the platform also emphasizes the financial dimension of AI infrastructure. Energy consumption directly affects operational costs, and companies often struggle to estimate the long-term financial impact of large-scale AI deployments. By translating technical metrics such as tokens, prompts, and compute cycles into monetary costs, the site makes the economic implications of AI infrastructure easier to understand.

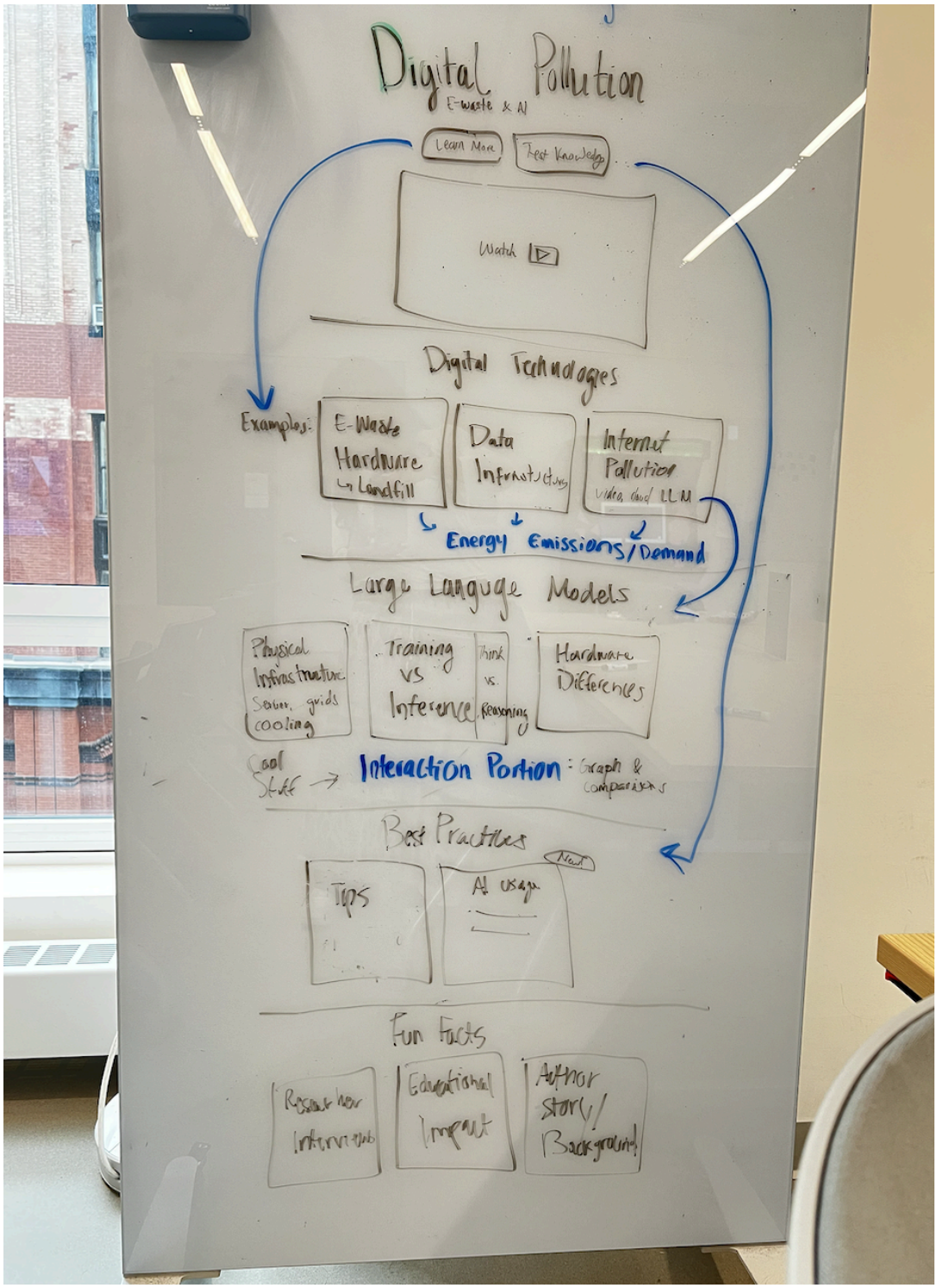
Ultimately, the goal of this platform is to create a space where users can explore how AI systems operate at scale, compare infrastructure options, and discover strategies for reducing both environmental impact and operational cost. By bringing together developers, companies, and sustainability-focused infrastructure providers, the site encourages greater transparency, collaboration, and community around building more energy-efficient AI systems.

Google Antigravity Download

https://vite.dev/

Cloudflare.com to deploy

Interactive Website: Visualizing AI's Energy Footprint



Insights from the benchmarking analysis will be translated into an interactive educational website built with HTML, CSS, and JavaScript. The site will visually simulate how a model processes a prompt, where

computation occurs, and how energy accumulates with repeated use. Rather than presenting energy use as an abstract statistic, the prototype is designed to let users see the consequences of everyday AI interactions and explore lower-energy alternatives in real time. Three core interactive features are planned:

- User interactions
    - Example: AI company → how much energy needed for ____ output, with this CPU, how many people can be on this, how many prompts, how long does it take to power this process, how much does it cost, where to get the power from
    - Linking companies and developers that build it. A platform to plug in info on what customers need, then connects to partners that provide better CPUs that don't require so much energy, or information on how to plan to use this.
        - There are a lot of tech sales, this site will bring more connections and community
- Financial aspects

Model Comparison Slider. Drawing on ML. For the ENERGY Leaderboard data, this feature will allow users to select and compare different models side by side to assess the energy cost per response. For example, a large reasoning model such as DeepSeek R1 consumes approximately 2.37 joules per token, while a smaller model such as GPT OSS 20B uses only around 0.028 joules per token [17]. Visualizing this contrast on a sliding scale makes the otherwise abstract concept of "model size" tangible and personally relevant.

Email vs. AI Calculator. This interactive form will let users explore the energy equivalences between familiar digital habits and AI use. For instance, if deleting 1,000 stored emails saves a measurable amount of server energy, how many AI prompts does that correspond to? Grounding AI energy costs in the familiar vocabulary of everyday digital pollution (unread emails, cloud storage) bridges the gap between abstract data and personal behavior.

Live Energy Meter Widget. Using the Zeus library in a Python-based cloud environment, a simulated "Live Energy Meter" will display the estimated energy cost of calling a model in real time. As users type a prompt, the widget will animate the accumulating energy cost token by token, making the forward-pass nature of inference visually intuitive. This widget is designed specifically to address the TDP Myth by displaying real measured values rather than theoretical maximums.

By pairing cloud-based experimentation with a public-facing prototype, this work bridges measurement and meaning. The goal is not only to quantify AI's energy footprint but also to translate technical results into an experience that encourages more mindful, informed, and sustainable engagement with AI systems.

Test websites:

- Figma: Output https://depth-spoof-51322959.figma.site/
- Code in Google Antigravity
- Implement Tokenizer - OpenAI API
- Launch on Cloudflare Pages

Broader Impacts

The environmental costs of digital technologies disproportionately affect low-income and developing regions, which often host the data infrastructure of wealthier nations. As Presberger and Bernauer (2023) note, this creates ecological inequality by outsourcing pollution while centralizing profit.

Raising awareness about LLM energy consumption contributes to ongoing discussions on AI ethics and sustainability. As Timnit Gebru and colleagues emphasize, unchecked model scaling threatens both social equity and environmental stability. By connecting ethical AI practices to measurable energy metrics, this project situates digital pollution within the movement for responsible innovation and climate justice. Its outcomes will encourage developers and consumers to make informed, sustainable choices in their digital practices.

If time permits, the project will also produce a short animated video inspired by TED-Ed that explains AI energy consumption, including the distinction between training and inference, the role of response verbosity and model selection, and concrete steps users can take to reduce their computational footprint, in an accessible format for general audiences.

Conclusion and Next Steps

Building on prior coursework in computer organization, data science, and environmental systems, this thesis unites technical, ethical, and educational approaches to address the hidden costs of computation. The final deliverables will include a research paper integrating cloud-based measurements, surveys, and literature findings; an interactive website visualizing the energy footprint of AI queries; and, if time permits, an educational animation illustrating AI's environmental impact and mitigation strategies.

To support the technical component, this project uses a cloud-computing grant of up to $1,000 in GPU credits for energy-use experiments on AWS or Google Cloud. Arya Boudaie, Software Engineer at Amazon, will serve as technical advisor, ensuring methodological accuracy and practical insight. Next steps include finalizing the advisory committee, designing the survey instrument, and developing the website prototype alongside initial cloud-based energy tests.

This project bridges research and education by transforming complex AI concepts into accessible tools for awareness. Making the environmental footprint of LLMs visible calls for a more responsible and sustainable approach to technological progress.

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